

Total No. of Questions : 8]

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Roll No

EC-803-GS
B.E. VIII Semester
Examination, December 2020
Grading System (GS)
Computer Networks
Time : Three Hours

Maximum Marks :70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain ISO-OSI reference model and list the major functions of all the layers.
b) Give detailed comparison of OSI and TCP/IP models.
2. What is CSMA scheme? Discuss non-persistent, 1-persistent and p-persistent CSMA with a diagram. Derive an expression for the through put of a slotted ALOHA system and compare with pure ALOHA system.
3. Explain Guided and Unguided Medium in detail.
4. a) Discuss the principle working of RFID.
b) Discuss the use of repeaters, hubs, bridges and routers in data link layer switching.
5. Discuss the advantages and disadvantages of credits versus sliding window protocols. Why does UDP exist? Would it not have been enough to just let user processes send raw IP packets?

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6. Explain Adaptive and Non-Adaptive Routing Algorithm.
7. How does selective-repeat-ARQ protocol differ from Go-back-N ARQ? Discuss with suitable diagrams. Name the three types of HDLC frames and give a brief explanation of each.
8. Write short notes on any two of the following
 - i) Collision free protocols
 - ii) 802.11, 802.16, Bluetooth
 - iii) IPv4 and IPv6 protocol
 - iv) Congestion control algorithm
 - v) Performance issues of the transport layer
